

Program 09: Organize kubernetes Pods using labels and namespace

Structure of Program:

- namespaces.yaml
- pods.yaml

Steps:

1. Create two files namespaces.yaml and pods.yaml

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ nano namespace.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ nano pod.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ cat namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ cat pod.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ █
```

2. Apply the YAML files

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ ls
namespace.yaml pod.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ kubectl apply -f namespace.yaml
namespace/dev created
namespace/prod created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ kubectl apply -f pod.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ █
```

3. Verify the pods

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
labeled-pod   1/1     Running   1 (4h44m ago)  7h47m
my-pod        1/1     Running   1 (4h44m ago)  7h53m
program-8     1/1     Running   2 (5m22s ago)  10d
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ █
```

4. Filter the pods with labels

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           2m52s
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           3m
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p9$ █
```